OligoTox-Hydrocephalus — Source and Provenance Register

NIH/NCATS Oligonucleotide Toxicity Open Data Challenge, Phase 2 · hydrocephalus and CSF-dynamics endpoints · supplementary to the four required submission documents

Every measurement in this release names the document it came from, and this register names every one of those documents: what database it was taken from, how it was retrieved, where in it the value sits, what may lawfully be done with it, and a link that resolves. 193 source records support 1361 measurement rows and 256 per-position chemistry rows across 53 compounds.

The register is generated from the data itself, never typed alongside it. Three checks in `qc/validate.py` enforce that every source carrying rows has a real citation, states its link, rights, evidence tier and retrieval route, and points at a document rather than a bare hostname. All 193 distinct URLs below were resolved on 2026-09-04; 193 returned a success status and none was dead.

How to trace any single number

Every row of `measurements.csv` carries four provenance columns, and together they resolve to one sentence in one document:

- **source_id** — the key of the row in `sources.csv`, which is the table reproduced in this document.
- **source_ref** — the stable external identifier: an NCT number, a PMID/PMCID/DOI, or a DailyMed setid with its publication date.
- **source_location** — where inside that document the value sits: a named results table, a LOINC-coded label section, an SmPC Annex I section number, a figure number.
- **evidence_quote** — for curated rows, the source's own words, so the reading is checkable without re-fetching the paper.

1 Databases and archives used

Six. Each is a primary public source; none of the release derives from a secondary compilation, a review article's table, or another curated dataset.

Database	What it supplies here	Records	Meas. rows	Pos. rows	Rights
ClinicalTrials.gov	Per-arm adverse-event counts with denominators, including reported zeros	160	786	0	US Government work / public domain
openFDA FAERS	Spontaneous-report counts by drug and MedDRA preferred term	1	456	0	US Government work / public domain
DailyMed (FDA SPL)	US label text, and the chemistry statements behind the modification maps	16	80	0	US Government work / public domain
PubMed Central / Europe PMC	Case reports, the disease background rate, and the nonclinical rows	7	31	138	CC BY 2.0; CC BY 4.0; CC BY-NC; CC BY-NC-ND; CC BY-NC-ND…
EMA EPAR	EU label text — the jurisdiction contrast	3	8	0	EMA publication
WHO INN lists	Sequence and per-position chemistry, by deterministic parse	6	0	118	WHO publication

Exact access route, per database

ClinicalTrials.gov. ClinicalTrials.gov REST API v2 — GET https://clinicaltrials.gov/api/v2/studies?query.intr=<drug> to enumerate, then GET /api/v2/studies/<NCT> for each record. Rows come from resultsSection.adverseEventsModule.

openFDA FAERS. openFDA drug/event REST API — GET https://api.fda.gov/drug/event.json?search=<drug> AND patient.reaction.reactionmeddrapt.exact:"<PT>", one exact query per (drug, term) pair.

DailyMed (FDA SPL). DailyMed REST API v2 — GET /dailymed/services/v2/spls/<setid>.xml. Each citation carries the setid and publication date of the exact label version read.

PubMed Central / Europe PMC. Europe PMC REST fullTextXML endpoint; open-access full text only, never a paywalled PDF.

EMA EPAR. European Public Assessment Report product-information PDFs from ema.europa.eu, text-extracted and swept section by section.

WHO INN lists. Recommended INN list PDFs from cdn.who.int, parsed residue by residue by scripts/parse_inn_sequences.py.

Raw payloads for every retrieval are kept under sources/raw/ so any extraction can be re-derived without re-querying the host.

2 Primary literature

Seven papers, all open access, all retrieved as full text through Europe PMC. These are the rows that carry a quotation, a figure number and an author attribution — the evidence a registry table cannot give.

Mönkkönen KS, Hakumäki JM, Hirst RA, et al. Intracerebroventricular antisense knockdown of G-alpha-i2 results in ciliary stasis and ventricular dilatation in the rat. BMC Neurosci. 2007;8:26.

DOI 10.1186/1471-2202-8-26 · PMID 17430589 · PMCID PMC1855344 · licence CC BY 2.0 · redistribution **cc_by** · 5 measurement rows, 54 position rows

https://pmc.ncbi.nlm.nih.gov/articles/PMC1855344/

The only source measuring one oligonucleotide BOTH in vitro (ciliary beat frequency in cultured ependymal cells) and in vivo (MRI ventricular volume), which is the extrapolation the Phase 2 brief calls a particular interest. Supplies three published sequences including two designed controls, and the only stated purity value in the release (HPLC-purified, 90-97%).

Guo J, Mi X, Zhan R, Li M, Wei L, Sun J. Aquaporin 4 Silencing Aggravates Hydrocephalus Induced by Injection of Autologous Blood in Rats. Med Sci Monit. 2018;24:4204-4212.

DOI 10.12659/MSM.906936 · PMID 29921834 · PMCID PMC6042309 · licence CC BY-NC · redistribution **cc_by_nc** · 3 measurement rows, 0 position rows

https://pmc.ncbi.nlm.nih.gov/articles/PMC6042309/

Toxic-direction nonclinical rows, and the dataset's only DESIGNED negative control (a scrambled non-targeting siRNA).

Stoker TB, Andresen KER, Barker RA. Hydrocephalus Complicating Intrathecal Antisense Oligonucleotide Therapy for Huntington's Disease. Mov Disord. 2021;36(1):263-264.

DOI 10.1002/mds.28359 · PMID 33125799 · PMCID PMC7894279 · licence CC BY 4.0 · redistribution **cc_by** · 5 measurement rows, 0 position rows

https://pmc.ncbi.nlm.nih.gov/articles/PMC7894279/

The index case for this endpoint: communicating hydrocephalus after intrathecal tominersen, attributed by the authors to a drug-induced sterile meningitis, resolved by ventriculoperitoneal shunting.

Viscidi E, Wang N, Juneja M, et al. The incidence of hydrocephalus among patients with and without spinal muscular atrophy (SMA): Results from a US electronic health records study. Orphanet J Rare Dis. 2021;16:207.

DOI 10.1186/s13023-021-01822-4 · PMID 33962637 · PMCID PMC8105953 · licence CC BY 4.0 · redistribution **cc_by** · 3 measurement rows, 0 position rows

https://pmc.ncbi.nlm.nih.gov/articles/PMC8105953/

THE CONFOUNDER CONTROL. Disease background rate of hydrocephalus in SMA over a study window that ends at nusinersen approval, so it is uncontaminated by the drug.

Lovett A, Chary S, Babu S, et al. Serious Neurologic Adverse Events in Tofersen Clinical Trials for Amyotrophic Lateral Sclerosis. Muscle Nerve. 2025;71:1006-1015.

DOI 10.1002/mus.28372 · PMID 40017137 · PMCID PMC12060635 · licence CC BY-NC-ND · redistribution **summary_stat_only** · 3 measurement rows, 0 position rows

https://pmc.ncbi.nlm.nih.gov/articles/PMC12060635/

ND licence term: only abstract-level summary statistics are carried; no underlying table is reproduced.

Wang Q, Xia X, Zhang H, et al. Targeting modulation of the choroid plexus blood-CSF barrier and CSF hypersecretion via lipid nanoparticle-mediated co-delivery of siRNA and resveratrol. Nat Commun. 2025;16:6389.

DOI 10.1038/s41467-025-61543-1 · PMID 40640139 · PMCID PMC12246246 · licence CC BY-NC-ND 4.0 · redistribution **summary_stat_only** · 2 measurement rows, 84 position rows

https://pmc.ncbi.nlm.nih.gov/articles/PMC12246246/

Supplies the only published sequences in this release (four SPAK siRNA duplexes, Methods/Materials) and the only protective-direction rows.

Nakayama T, El Achkar CM, Burbano LE, et al. Antisense oligonucleotide-mediated knockdown therapy in two infants with severe KCNT1 epileptic encephalopathy. Nat Med. 2026;32:1411.

DOI 10.1038/s41591-026-04314-9 · PMID 41981306 · PMCID PMC13099374 · licence CC BY-NC-ND 4.0 · redistribution **summary_stat_only** · 10 measurement rows, 0 position rows

https://pmc.ncbi.nlm.nih.gov/articles/PMC13099374/

The SECOND independent drug-attributed signal, and the reason this release does not call hydrocephalus a tominersen-specific finding: a different ASO (valeriasen/KT777), a different target, indication and age group, and both treated infants developed the endpoint. ND licence term: summary statistics only.

3 Regulatory documents

3.1 US labels — DailyMed Structured Product Labels

A US label is revised in place, so a link to the drug's DailyMed page does not identify what was read. Each row below carries the setid and the publication date of the specific label version parsed. A label that is silent on the endpoint produces an explicit `measured_null` row rather than no row at all, which is why silent labels appear here.

Drug	Label version read	Rows	DailyMed setid
casimersen	Feb 25, 2026	5	e9e5fd44-eeda-4580-bba1-a734828bbcc3
defibrotide	Jul 04, 2025	5	2c3db989-d7ad-41ed-9ebf-698dcf6c24ec
eplontersen	Apr 30, 2026	5	d7dcb847-71dd-4fff-82d0-d43a465fc096
eteplirsen	Aug 20, 2025	5	33bff678-7829-479e-9110-b8e33a0bc0aa
givosiran	Sep 22, 2025	5	167e663c-11e1-497b-a3fc-951d65d58eaa
golodirsen	Jul 08, 2025	5	35c227d1-5b24-44b0-b5d3-f0f6b1c46bd5
imetelstat	Apr 27, 2026	5	b0fab7ca-e578-43c5-9df6-bdaff4182257
inclisiran	Aug 21, 2026	5	6fc0afca-4513-4c35-b594-6544aee29a44
inotersen	Nov 17, 2025	5	3d31f4c0-35bb-c990-e063-6394a90a749f
lumasiran	Dec 15, 2025	5	16985a31-f5e4-4557-9266-fc78d4bc5055
nedosiran	Apr 10, 2025	5	ace9d4bc-4d20-4beb-9e9d-888690424833
nusinersen	Apr 06, 2026	5	dd70cd5f-b0fc-4ba4-a5ea-89a34778bd94
patisiran	Sep 29, 2025	5	e87ec36f-b4b4-49d4-aea4-d4ffb09b0970
tofersen	Nov 18, 2024	5	81356b45-1cb7-4eef-88ea-e44cc18b47c5
viltolarsen	Jul 21, 2025	5	1ffff9a8-6d6a-4dcb-8493-1b6cc3a5d123
vutrisiran	Nov 17, 2025	5	8db0facb-81b6-4006-9239-27dc6409c5d3

3.2 EU labels — EMA EPAR product information

Included because the two regulators reached materially different positions on the same molecules: the EMA gives hydrocephalus its own subheading under section 4.4 for nusinersen, where the US label mentions it only under postmarketing experience. A jurisdiction contrast on an identical molecule is itself a datum about how strongly the signal is judged.

European Medicines Agency. Qalsody (tofersen) EPAR product information, Annex I Summary of Product Characteristics (first authorised 29 May 2024).

https://www.ema.europa.eu/en/documents/product-information/qalsody-epar-product-information_en.pdf · 5 rows · redistribution **verify**

European Medicines Agency. Spinraza (nusinersen) EPAR product information, Annex I Summary of Product Characteristics (first authorised 30 May 2017, latest renewal 31 January 2022).

https://www.ema.europa.eu/en/documents/product-information/spinraza-epar-product-information_en.pdf · 2 rows · redistribution **verify**

European Medicines Agency. Tegsedi (inotersen) EPAR product information, Annex I Summary of Product Characteristics (first authorised 06 July 2018, latest renewal 24 March 2023).

https://www.ema.europa.eu/en/documents/product-information/tegsedi-epar-product-information_en.pdf · 1 row · redistribution **verify**

EMA reuse terms were not established in this session, so every EU-label row is marked `redistribution = verify` rather than assumed open. A redistributor should resolve the licence before republishing those values.

4 Sequence and chemistry provenance — WHO INN lists

No US label prints an oligonucleotide's sequence. The WHO Recommended INN entry does, indirectly but completely: it spells out every residue longhand — its sugar, its base, any 5-methylation, and whether the linkage to the next residue is a phosphorothioate or a plain phosphodiester — so both the sequence and the per-position modification map are recovered by deterministic parse rather than judgement.

INN list	Compounds parsed from it	Position rows	Link
List 74	nusinersen	18	rl74.pdf
List 75	volanesorsen	20	rl75.pdf
List 77	inotersen	20	rl77.pdf
List 81	tofersen	20	rl81.pdf
List 83	tominersen	20	rl83.pdf
List 85	eplontersen	20	rl85.pdf

The parse is validated against evidence that does not come from the INN list, and `scripts/parse_inn_sequences.py` refuses to emit a sequence that disagrees: parsed length must equal the length derived from the label's molecular formula, and parsed phosphorothioate and phosphodiester counts must equal the label's own statement where it makes one. Two further lists were parsed successfully but contribute nothing to this release: zorevunersen (list 87) and elsunersen (list 92) have no trial with posted results and therefore no measurement row for a sequence to attach to. Their parses are retained in `data/inn_sequences.json`.

5 Pharmacovigilance — openFDA FAERS

FDA Adverse Event Reporting System (FAERS), queried through the openFDA drug/event API — https://api.fda.gov/drug/event.json · 456 measurement rows over 19 compounds.

One exact query per (drug, MedDRA preferred term) pair, over 28 preferred terms spanning both endpoint tiers. Two design decisions are worth stating because both were corrections to earlier defects. First, an aggregated `count` query truncates at 100 buckets and silently loses the tail, so exact per-pair queries replaced it. Second, term strings are now verified against FAERS itself before use: `CEREBROSPINAL FLUID PROTEIN INCREASED` matches nothing because FAERS stores that preferred term as `CSF PROTEIN INCREASED`, and the unverified spelling had recorded a false zero for all 19 drugs. A pre-flight vocabulary probe now drops any term string the database does not know, so the same class of error cannot recur silently.

Spontaneous reports are counts of reports, not patients, and carry no exposure denominator. No disproportionality statistic is computed from them anywhere in this release.

6 Clinical trial registry — the complete list

161 registered trials of an oligonucleotide therapeutic with posted results, every one of them enumerated by query rather than chosen. This matters more than any single citation: the first version of this component carried a hand-written map of 23 trials — the ones that surfaced while chasing the signal — which biases a dataset toward the compounds already suspected and silently discards the negative evidence that makes a positive interpretable. Asking the registry for all of them is what makes the 755 reported zeros in this release meaningful.

Route is taken from each trial's own intervention and arm text, never assumed from the compound, and the matched sentence is stored in `route_evidence` so the assignment is checkable. Distribution: subcutaneous 58, intravenous 46, NOT_REPORTED 20, intrathecal_lumbar 19, intravitreal 16, oral 1, topical_enema 1.

An INN-only query is not sufficient, and the failure is silent. The tominersen first-in-human trial NCT02519036 posts a full adverse-event module, but its record never contains the string “tominersen” — the drug is registered as ISIS 443139 with other name IONIS HTTRx — so no query on the INN could reach it and it was missing from the first registry. Trials are now also reached through development codes that a record equates with a compound inside a single intervention entry, and each such trial carries the record that supplied the alias in its `discovery` column (3 trials). A coverage sweep reports any fetched trial that posts adverse events but reaches no compound, so the next gap of this kind is loud rather than silent.

NCT	Compound	Route	Status	Title
NCT03186989	BIIB080	intrathecal lumbar	Completed	Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of…
NCT04494256	BIIB105	intrathecal lumbar	Terminated	A Study to Assess the Safety, Tolerability, and Effect on Disease…
NCT05032196	WVE-003	NOT REPORTED	Completed	Study of WVE-003 in Patients With Huntington's Disease
NCT03225833	WVE-120101	NOT REPORTED	Terminated	Safety and Tolerability of WVE-120101 in Patients With Huntington's…
NCT04617847	WVE-120101	intrathecal lumbar	Terminated	Open-label Extension Study to Evaluate the Safety and Tolerability of…
NCT03225846	WVE-120102	NOT REPORTED	Terminated	Safety and Tolerability of WVE-120102 in Patients With Huntington's…
NCT04617860	WVE-120102	intrathecal lumbar	Terminated	Open-label Extension Study to Evaluate the Safety and Tolerability of…
NCT02525523	alicaforsen	topical enema	Completed	Efficacy of Alicaforsen in Pouchitis Patients Who Have Failed to…
NCT01681433	apatorsen	intravenous	Terminated	OGX-427 in Metastatic Castrate-Resistant Prostate Cancer With…
NCT01780545	apatorsen	intravenous	Completed	Phase 2 Study of Docetaxel +/- OGX-427 in Patients With Relapsed or…
NCT01829113	apatorsen	intravenous	Completed	Phase II Trial of Carboplatin and Pemetrexed +/- OGX-427 in Untreated…
NCT01844817	apatorsen	intravenous	Completed	Phase II Trial Of Gemcitabine Plus Nab-Paclitaxel +/- OGX-427 In…
NCT05330455	bepirovirsen	subcutaneous	Terminated	Study of GSK3965193 in Healthy Participants and Participants Living…
NCT06058390	bepirovirsen	NOT REPORTED	Completed	A Study to Evaluate the Relative Bioavailability of Subcutaneous…

NCT06422767	bepirovirsen	subcutaneous	Completed	Study to Evaluate the Effect of Bepirovirsen on Cardiac Conduction as…
NCT02530905	casimersen	intravenous	Completed	Dose-Titration and Open-label Extension Study of SRP-4045 in Advanced…
NCT04179409	casimersen; eteplirsen;…	NOT REPORTED	Completed	A 48-Week, Open Label, Study to Evaluate the Efficacy and Safety of…
NCT03532542	casimersen; golodirsen	intravenous	Terminated	An Extension Study to Evaluate Casimersen or Golodirsen in Patients…
NCT00138658	custirsen	intravenous	Completed	A Study of OGX-011/Gemcitabine/Platinum-Based Regimen in Stage IIIB/IV…
NCT00327340	custirsen	intravenous	Completed	Evaluation of Safety and Feasibility of OGX-011 in Combination With…
NCT01839604	danvatirsen	intravenous	Completed	A Phase I/Ib Study of AZD9150 (ISIS-STAT3Rx) in Patients With…
NCT02417753	danvatirsen	intravenous	Terminated	AZD9150, a STAT3 Antisense Oligonucleotide, in People With Malignant…
NCT02499328	danvatirsen	NOT REPORTED	Completed	Study to Assess MEDI4736 With Either AZD9150 or AZD5069 in Advanced…
NCT03334617	danvatirsen	intravenous	Active Not Recruiting	Phase II Umbrella Study of Novel Anti-cancer Agents in Participants…
NCT03794544	danvatirsen	intravenous	Completed	Neoadjuvant Durvalumab Alone or in Combination With Novel Agents in…
NCT05814666	danvatirsen	intravenous	Terminated	Activity and Safety of Danvatirsen and Pembrolizumab in HNSCC
NCT00358501	defibrotide	intravenous	Completed	Defibrotide for the Treatment of Severe Hepatic Veno-Occlusive Disease…
NCT00628498	defibrotide	intravenous	Completed	Defibrotide for Patients With Hepatic Veno-occlusive Disease: A…
NCT02851407	defibrotide	intravenous	Completed	Study Comparing Efficacy and Safety of Defibrotide vs Best Supportive…
NCT02876601	defibrotide	intravenous	Completed	Defibrotide in the Human Endotoxemia Model --- an Exploratory Trial…
NCT03339297	defibrotide	intravenous	Completed	An Open-Label Study of Defibrotide for the Prevention of Acute…
NCT03384693	defibrotide	intravenous	Completed	Defibrotide TMA Prophylaxis Pilot Trial
NCT03954106	defibrotide	NOT REPORTED	Terminated	A Safety and Efficacy Study of Defibrotide in the Prevention of…
NCT04530604	defibrotide	intravenous	Completed	Defibrotide Therapy for SARS-CoV2 (COVID-19) Acute Respiratory Distress…
NCT04030598	donidalorsen	subcutaneous	Completed	A Study to Assess the Clinical Efficacy of Donidalorsen (Also Known as…
NCT04307381	donidalorsen	subcutaneous	Completed	An Extension Study of Donidalorsen (IONIS-PKK-LRx) in Participants With…
NCT05139810	donidalorsen	subcutaneous	Completed	OASIS-HAE: A Study to Evaluate the Safety and Efficacy of Donidalorsen…

NCT03728634	eplontersen	subcutaneous	Completed	Evaluate the Safety and Tolerability, as Well as the Pharmacokinetic…
NCT04136184	eplontersen; inotersen	subcutaneous	Completed	NEURO-TTRansform: A Study to Evaluate the Efficacy and Safety of…
NCT00159250	eteplirsen	NOT REPORTED	Completed	Safety and Efficacy Study of Antisense Oligonucleotides in Duchenne…
NCT00844597	eteplirsen	intravenous	Completed	Dose-Ranging Study of AVI-4658 to Induce Dystrophin Expression in…
NCT01396239	eteplirsen	intravenous	Completed	Efficacy Study of AVI-4658 to Induce Dystrophin Expression in Selected…
NCT01540409	eteplirsen	intravenous	Completed	Efficacy, Safety, and Tolerability Rollover Study of Eteplirsen in…
NCT02255552	eteplirsen	intravenous	Completed	Study of Eteplirsen in DMD Patients
NCT02286947	eteplirsen	intravenous	Completed	Safety Study of Eteplirsen to Treat Advanced Stage Duchenne Muscular…
NCT02420379	eteplirsen	intravenous	Completed	Safety Study of Eteplirsen to Treat Early Stage Duchenne Muscular…
NCT03218995	eteplirsen	intravenous	Completed	Study of Eteplirsen in Young Participants With Duchenne Muscular…
NCT03985878	eteplirsen	intravenous	Terminated	A Study to Evaluate Safety, Tolerability, and Efficacy of Eteplirsen in…
NCT02554773	fitusiran	subcutaneous	Completed	An Open-label Extension Study of an Investigational Drug, Fitusiran, in…
NCT03417102	fitusiran	subcutaneous	Completed	A Study of Fitusiran (ALN-AT3SC) in Severe Hemophilia A and B Patients…
NCT03417245	fitusiran	subcutaneous	Completed	A Study of Fitusiran (ALN-AT3SC) in Severe Hemophilia A and B Patients…
NCT03549871	fitusiran	subcutaneous	Completed	A Study of Fitusiran in Severe Hemophilia A and B Patients Previously…
NCT02949830	givosiran	subcutaneous	Completed	A Study to Evaluate Long-term Safety and Clinical Activity of Givosiran…
NCT03338816	givosiran	subcutaneous	Completed	ENVISION: A Study to Evaluate the Efficacy and Safety of Givosiran…
NCT02310906	golodirsen	intravenous	Completed	Phase I/II Study of SRP-4053 in DMD Patients
NCT00094835	imetelstat	intravenous	Completed	Study to Evaluate Motesanib With or Without Carboplatin/Paclitaxel or…
NCT00101907	imetelstat	intravenous	Terminated	Safety of AMG 706 Plus Panitumumab Plus Gemcitabine-Cisplatin in the…
NCT00427349	imetelstat	oral	Completed	AMG 706 and Octreotide in Treating Patients With Low-Grade…
NCT00574951	imetelstat	NOT REPORTED	Terminated	AMG 706 in Treating Patients With Persistent or Recurrent Ovarian…
NCT01731951	imetelstat	intravenous	Completed	Imetelstat Sodium in Treating Participants With Primary or Secondary…
NCT01836549	imetelstat	intravenous	Terminated	Imetelstat Sodium in Treating Younger Patients With Recurrent or…

NCT02426086	imetelstat	intravenous	Completed	Study to Evaluate Activity of 2 Dose Levels of Imetelstat in…
NCT02598661	imetelstat	intravenous	Active Not Recruiting	Study to Evaluate Imetelstat (GRN163L) in Participants With…
NCT05583552	imetelstat	intravenous	Active Not Recruiting	Study to Evaluate Imetelstat in Patients With High-Risk MDS or AML…
NCT02597127	inclisiran	subcutaneous	Completed	Trial to Evaluate the Effect of ALN-PCSSC Treatment on Low Density…
NCT02963311	inclisiran	subcutaneous	Completed	A Study of ALN-PCSSC in Participants With Homozygous Familial…
NCT03060577	inclisiran	subcutaneous	Completed	An Extension Trial of Inclisiran in Participants With Cardiovascular…
NCT03397121	inclisiran	subcutaneous	Completed	Trial to Evaluate the Effect of Inclisiran Treatment on Low Density…
NCT03399370	inclisiran	subcutaneous	Completed	Inclisiran for Participants With Atherosclerotic Cardiovascular Disease…
NCT03400800	inclisiran	subcutaneous	Completed	Inclisiran for Subjects With ASCVD or ASCVD-Risk Equivalents and…
NCT03814187	inclisiran	subcutaneous	Completed	Trial to Assess the Effect of Long Term Dosing of Inclisiran in…
NCT03851705	inclisiran	subcutaneous	Completed	A Study of Inclisiran in Participants With Homozygous Familial…
NCT04652726	inclisiran	subcutaneous	Completed	Study to Evaluate Efficacy and Safety of Inclisiran in Adolescents With…
NCT04659863	inclisiran	subcutaneous	Completed	Study to Evaluate Efficacy and Safety of Inclisiran in Adolescents With…
NCT04666298	inclisiran	subcutaneous	Completed	Study of Efficacy and Safety of Inclisiran in Japanese Participants…
NCT04807400	inclisiran	subcutaneous	Completed	Study in Primary Care Evaluating Inclisiran Delivery Implementation +…
NCT04873934	inclisiran	NOT REPORTED	Completed	Management of LDL-cholesterol With Inclisiran + Usual Care Compared to…
NCT04929249	inclisiran	NOT REPORTED	Completed	A Randomized Study to Evaluate the Effect of an "Inclisiran First"…
NCT05192941	inclisiran	subcutaneous	Completed	Study of Efficacy, Safety, Tolerability and Quality of Life of…
NCT05763875	inclisiran	subcutaneous	Completed	Efficacy and Safety of Inclisiran as Monotherapy in Patients With…
NCT05888103	inclisiran	NOT REPORTED	Completed	Efficacy and Safety of Inclisiran as Monotherapy in Chinese Adults With…
NCT01737398	inotersen	subcutaneous	Completed	Efficacy and Safety of Inotersen in Familial Amyloid Polyneuropathy
NCT02175004	inotersen	subcutaneous	Completed	Extension Study Assessing Long Term Safety and Efficacy of IONIS-TTR Rx…
NCT04306510	inotersen	subcutaneous	Terminated	A Study to Characterize Adverse Events Occurring Within One Day of…
NCT02706886	lumasiran	subcutaneous	Completed	Study of Lumasiran in Healthy Adults and Patients With Primary…

NCT03350451	lumasiran	subcutaneous	Completed	An Extension Study of an Investigational Drug, Lumasiran (ALN-GO1), in…
NCT03681184	lumasiran	subcutaneous	Completed	A Study to Evaluate Lumasiran in Children and Adults With Primary…
NCT03905694	lumasiran	subcutaneous	Completed	A Study of Lumasiran in Infants and Young Children With Primary…
NCT04152200	lumasiran	subcutaneous	Completed	A Study to Evaluate Lumasiran in Patients With Advanced Primary…
NCT05161936	lumasiran	subcutaneous	Terminated	A Study to Evaluate Lumasiran in Adults With Recurrent Calcium Oxalate…
NCT00362180	mipomersen	subcutaneous	Completed	Measure Liver Fat Content After ISIS 301012 (Mipomersen) Administration
NCT00477594	mipomersen	subcutaneous	Completed	Open Label Extension of ISIS 301012 (Mipomersen) to Treat Familial…
NCT00607373	mipomersen	subcutaneous	Completed	Study to Assess the Safety and Efficacy of ISIS 301012 (Mipomersen) in…
NCT00694109	mipomersen	subcutaneous	Completed	An Open-label Extension Study to Assess the Long-term Safety and…
NCT00706849	mipomersen	subcutaneous	Completed	Efficacy and Safety Study of ISIS 301012 (Mipomersen) as Add-on in…
NCT00707746	mipomersen	subcutaneous	Completed	Safety and Efficacy Study of ISIS 301012 (Mipomersen) Administration in…
NCT00770146	mipomersen	subcutaneous	Completed	Safety and Efficacy of Mipomersen (ISIS 301012) As Add-on Therapy in…
NCT00794664	mipomersen	subcutaneous	Completed	Safety and Efficacy of Mipomersen in Patients With Severe…
NCT01475825	mipomersen	subcutaneous	Completed	A Study of the Safety and Efficacy of Two Different Regimens of…
NCT03847909	nedosiran	subcutaneous	Completed	A Study to Evaluate DCR-PHXC in Children and Adults With Primary…
NCT05001269	nedosiran	subcutaneous	Completed	Nedosiran in Pediatric Patients From Birth to 11 Years of Age With PH…
NCT01703988	nusinersen	intrathecal lumbar	Completed	An Open-label Safety, Tolerability and Dose-Range Finding Study of…
NCT01839656	nusinersen	intrathecal lumbar	Completed	A Study to Assess the Efficacy, Safety and Pharmacokinetics of…
NCT02193074	nusinersen	intrathecal lumbar	Terminated	A Study to Assess the Efficacy and Safety of Nusinersen (ISIS 396443)…
NCT02292537	nusinersen	intrathecal lumbar	Completed	A Study to Assess the Efficacy and Safety of Nusinersen (ISIS 396443)…
NCT02386553	nusinersen	intrathecal lumbar	Completed	A Study of Multiple Doses of Nusinersen (ISIS 396443) Delivered to…
NCT02462759	nusinersen	intrathecal lumbar	Terminated	A Study to Assess the Safety and Tolerability of Nusinersen (ISIS…
NCT02594124	nusinersen	intrathecal lumbar	Completed	A Study for Participants With Spinal Muscular Atrophy (SMA) Who…

NCT04089566	nusinersen	intrathecal lumbar	Completed	Study of Nusinersen (BIIB058) in Participants With Spinal Muscular…
NCT05386680	nusinersen	intrathecal lumbar	Completed	Phase IIIb, Open-label, Multi-center Study to Evaluate Safety,…
NCT03385239	olezarsen	subcutaneous	Completed	Study of ISIS 678354 (AKCEA-APOCIII-LRx) in Participants With…
NCT04568434	olezarsen	subcutaneous	Completed	A Study of Olezarsen (Formerly Known as AKCEA-APOCIII-LRx) Administered…
NCT05355402	olezarsen	subcutaneous	Completed	A Study of Olezarsen (Formerly Known as AKCEA-APOCIII-LRx) in Adults…
NCT01617967	patisiran	intravenous	Completed	Safety and Tolerability of Patisiran (ALN-TTR02) in Transthyretin (TTR)…
NCT01960348	patisiran	intravenous	Completed	APOLLO: The Study of an Investigational Drug, Patisiran (ALN-TTR02),…
NCT01961921	patisiran	intravenous	Completed	The Study of ALN-TTR02 (Patisiran) for the Treatment of Transthyretin…
NCT02510261	patisiran	intravenous	Completed	The Study of an Investigational Drug, Patisiran (ALN-TTR02), for the…
NCT03862807	patisiran	intravenous	Completed	Patisiran in Patients With Hereditary Transthyretin-mediated…
NCT03997383	patisiran	intravenous	Completed	APOLLO-B: A Study to Evaluate Patisiran in Participants With…
NCT03759379	patisiran; vutrisiran	subcutaneous	Completed	HELIOS-A: A Study of Vutrisiran (ALN-TTRSC02) in Patients With…
NCT05635045	patisiran; vutrisiran	NOT REPORTED	Completed	Evuzamitide in PET/CT to Measure Potential Therapeutic Response in ATTR
NCT00239928	pegaptanib	NOT REPORTED	Completed	Clinical Study Of Pegaptanib Sodium (EYE001) For Wet-Type Age-Related…
NCT00242580	pegaptanib	intravitreal	Completed	A Safety and Efficacy Study Comparing the Combination Treatments of…
NCT00324116	pegaptanib	intravitreal	Completed	Evaluation Of Safety And Efficacy Of 0.3 Mg/Eye Macugen In Patients…
NCT00327470	pegaptanib	NOT REPORTED	Terminated	An Open Label Trial to Investigate Macugen for the Preservation of…
NCT00406107	pegaptanib	intravitreal	Completed	Evaluation of Macugen Treatment of Macular Edema Due to Branch Retinal…
NCT00460408	pegaptanib	intravitreal	Completed	Ocular Safety Of Patients Receiving Macugen For Neovascular Age Related…
NCT00549055	pegaptanib	intravitreal	Completed	A Study On The Efficacy Of Macugen Injections In Patients With…
NCT00605280	pegaptanib	intravitreal	Completed	A Multi-Center Trial To Evaluate The Safety And Efficacy Of Pegaptanib…
NCT00735943	pegaptanib	NOT REPORTED	Terminated	Macugen Observational Study
NCT00787319	pegaptanib	NOT REPORTED	Completed	Long-Term Non-Interventional Study (NIS) To Investigate The Safety And…

NCT00790803	pegaptanib	intravitreal	Completed	Pegaptanib Therapy in Non-Infectious Uveitic Cystoid Macular Edema
NCT00858208	pegaptanib	intravitreal	Completed	Efficacy And Safety Of Macugen In Patients With Neovascular AMD In…
NCT01100307	pegaptanib	intravitreal	Completed	A Phase 3 Study To Compare The Efficacy And Safety Of 0.3 MG Pegaptanib…
NCT01189461	pegaptanib	NOT REPORTED	Completed	Study To Evaluate Safety And Tolerability Of Pegaptanib Sodium In…
NCT01245387	pegaptanib	intravitreal	Completed	Observational Study Of The Long-Term Effect Of Macugen In Patients With…
NCT01486238	pegaptanib	intravitreal	Completed	Pegaptanib for Retinal Edema Secondary to Diabetic Vascular…
NCT01573572	pegaptanib	NOT REPORTED	Completed	Extension Study to Gather Data on Effect of Macugen on the Corneal…
NCT03140969	sepofarsen	intravitreal	Completed	Study to Evaluate QR-110 in Leber's Congenital Amaurosis (LCA) Due to…
NCT03913130	sepofarsen	intravitreal	Terminated	Extension Study to Study PQ-110-001 (NCT03140969)
NCT02623699	tofersen	intrathecal lumbar	Completed	An Efficacy, Safety, Tolerability, Pharmacokinetics and…
NCT03070119	tofersen	NOT REPORTED	Completed	Long-Term Evaluation of BIIB067 (Tofersen)
NCT02519036	tominersen	intrathecal lumbar	Completed	Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of ISIS…
NCT03342053	tominersen	intrathecal lumbar	Completed	A Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and…
NCT03761849	tominersen	intrathecal lumbar	Completed	A Study to Evaluate the Efficacy and Safety of Intrathecally…
NCT03842969	tominersen	intrathecal lumbar	Completed	An Open-Label Extension Study to Evaluate Long-Term Safety and…
NCT04000594	tominersen	intrathecal lumbar	Completed	A Study to Investigate the Pharmacokinetics and Pharmacodynamics of…
NCT00761280	trabedersen	subcutaneous	Terminated	Efficacy and Safety of AP 12009 in Patients With Recurrent or…
NCT05085964	ultevursen	intravitreal	Terminated	An Open-Label Extension Study to Evaluate Safety & Tolerability of…
NCT05158296	ultevursen	intravitreal	Terminated	Study to Evaluate the Efficacy Safety and Tolerability of Ultevursen in…
NCT05176717	ultevursen	intravitreal	Terminated	Study to Evaluate the Efficacy Safety and Tolerability of QR-421a in…
NCT02740972	viltolarsen	intravenous	Completed	Safety and Dose Finding Study of NS-065/NCNP-01 in Boys With Duchenne…
NCT03167255	viltolarsen	intravenous	Completed	Extension Study of NS-065/NCNP-01 in Boys With Duchenne Muscular…
NCT04060199	viltolarsen	intravenous	Completed	Study to Assess the Efficacy and Safety of Viltolarsen in Ambulant Boys…

NCT04956289	viltolarsen	intravenous	Completed	Study to Assess the Safety, Tolerability, and Efficacy of Viltolarsen…
NCT02211209	volanesorsen	subcutaneous	Completed	The APPROACH Study: A Study of Volanesorsen (Formerly IONIS-APOCIIIRx)…
NCT02300233	volanesorsen	subcutaneous	Completed	The COMPASS Study: A Study of Volanesorsen (Formally ISIS-APOCIIIRx) in…
NCT02527343	volanesorsen	subcutaneous	Terminated	The BROADEN Study: A Study of Volanesorsen (Formerly IONIS-APOCIIIRx)…
NCT02639286	volanesorsen	subcutaneous	Completed	Efficacy, Safety and Tolerability of ISIS 304801 in People With Partial…
NCT02658175	volanesorsen	subcutaneous	Completed	The Approach Open Label Study: A Study of Volanesorsen (Formerly…
NCT04153149	vutrisiran	subcutaneous	Active Not Recruiting	HELIOS-B: A Study to Evaluate Vutrisiran in Patients With Transthyretin…

7 Locator verification

A provenance register is only as good as its locators, so every URL in it is resolved and the status recorded in `notes/link_check.json` by `scripts/check_source_links.py`. On 2026-09-04, 193 distinct URLs were checked: 193 ok. A 403 or 429 would be recorded as *blocked* (the host answered but refused the request) rather than dead, and a 404 as *dead*, which fails the register.

8 What was deliberately not used

An exclusion that is not written down is indistinguishable from an oversight. Each of these is enforced in code at the location named.

Excluded	Reason	Where enforced
Viral-vector constructs (AAV, lentiviral shRNA)	Gene-therapy vectors, not oligonucleotide therapeutics. This cost the closest thing to a human in vitro finding — a choroid-plexus shRNA study — and is a curation judgement a reviewer may reasonably want to revisit, not a mechanical rule.	curation rule, applied at source triage
imlifidase	Matched an oligonucleotide drug-name query but is an IgG-degrading endopeptidase. Recorded by name with the reason rather than silently dropped.	`discover_ctgov_trials.py` NOT_OLIGONUCLEOTIDE
Four morpholinos, from the position table only	Phosphorus count is P = n for eteplirsen, golodirsen and casimersen but P = n–1 for viltolarsen, because some carry a 5'-piperazine bearing an extra phosphorus. Length is therefore ambiguous, and a per-position table cannot rest on an ambiguous length. They contribute measurement rows, just no position rows.	`build_modifications.py` EXCLUDED
Duplex siRNAs, from the INN parse only	The INN entry names both strands; recovering them needs a strand convention and a duplex reverse-complement check this parser has not been validated for. The parser refuses a duplex loudly rather than silently returning one strand of it.	`parse_inn_sequences.py` duplex guard
valeriasen sequence and chemistry	Published in the source's Extended Data Table 1 as an image whose bold/underline 2'-MOE encoding does not survive text extraction. Not transcribed by eye.	`build_modifications.py` EXCLUDED
Numbers read off figures	Both nonclinical sources publish their ventricular measurements graphically. Those rows carry readout_value NOT_REPORTED and readout_is_qualitative TRUE rather than a digitised estimate.	`build_nonclinical.py`

9 Retrieved and verified, but not extracted

A source-discovery pass returned 100 further sources that were retrieved and inspected but do not carry a `source_id` in this release. They are listed so the gap is visible rather than silent. Their absence is a completeness limit, not a quality one: nothing in `data/` depends on any of them. `notes/source_backlog.csv` carries the full list with retrieval routes, exact loci and per-source caveats.

Evidence type	Sources	Est. rows	Highest-value example
clinical trial AE	26	160	EudraCT 2020-005193-94 / NCT04931862 — FOCUS-C9 (WVE-004-001): Phase 1b/2a…
pharmacovigilance	21	627	EudraVigilance DAP — per-PT outcome, reporter-qualification and age/sex breakdowns…
regulatory label	17	96	Qalsody (tofersen) — CHMP Assessment report, EPAR public assessment report
nonclinical invivo	17	94	NDA 206488 EXONDYS 51 (eteplirsen) Pharmacology/Toxicology NDA Review and Evaluation,…
background incidence	7	68	The incidence of hydrocephalus among patients with and without spinal muscular…
clinical case report	6	24	Antisense oligonucleotide-mediated knockdown therapy in two infants with severe KCNT1…
review mechanism	5	2	Potential disease-modifying therapies for Huntington's disease: lessons learned and…
therapeutic direction	1	5	Targeting modulation of the choroid plexus blood-CSF barrier and CSF hypersecretion…

The single highest-value missing modality is the EMA EPAR *assessment reports* (scientific discussion and risk-management plan), as distinct from the EPAR *product information* that this release does carry. Those are where a regulator states an adjudicated causal judgement and a date. Also entirely absent: EudraVigilance, WHO VigiBase, the FDA pharmacology/toxicology review packages, PMDA documents, the EU Clinical Trials Register, and patent literature. Every spontaneous-report row here comes from one database, openFDA FAERS, and the nusinersen signal was first acted on in Europe.

Row-count estimates are the discovery pass's own and are not verified here; they also double-count wherever a backlog entry overlaps rows already present. Nothing in this section is evidence for any claim — it is a work list.

10 Rights, per source

Redistribution terms are tracked per row, not per dataset, so lawful reuse is verifiable at the record level. The curation layer — the schema, the grading rubric, the derived columns — is offered under CC BY 4.0; the underlying source rights are as follows and are not altered by that grant.

Redistribution term	What it permits	Sources	Measurement rows
public_domain	Values may be reproduced freely (US Government works)	177	1322
verify	Resolve the licence with the publisher before republishing values	9	8
summary_stat_only	Summary statistics only; no underlying table reproduced (ND term)	3	15
cc_by	Reproduce with attribution	3	13
cc_by_nc	Reproduce with attribution, non-commercial only	1	3

No source in this release contains PII, PHI or individual-level human-subjects data, so there is no privacy or consent barrier to sharing any of it.

Generated from data/sources.csv, data/trial_registry.csv, notes/link_check.json, notes/source_backlog.csv and qc/stats.json by docs/build_sources_pdf.py. 193 sources · 1361 measurements · 53 compounds · 53 QC checks passing.